

Supplementary Data for FruitCropXL: a modular, configuration-driven functional–structural modelling framework for perennial fruit crops

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Supplementary Methods

This supplement provides configuration material and equations supporting the main Methods. Section 1 introduces the principal configuration dimensions and commented example files; Section 2 presents the additional physiological equations and implementation details; Section 3 documents software structure and verification; Section 4 presents the compound dry-to-wet forcing and controlled-environment grapevine settings and analyses; Section 5 describes development, growth, management and state continuation; and Section 6 consolidates notation.

1 Configuration dimensions and commented example files

The principal configuration dimensions, their user-facing controls and their appropriate interpretation are summarised in Table S1.

Table S1: Principal configuration dimensions, their user-facing controls and appropriate interpretation.

Dimension	Principal controls and choices	Appropriate use and interpretation
Crop template and organ geometry	<code>species</code> , <code>leafShapeSpecies</code> ; crop parameters and grapevine, apple or kiwifruit leaf meshes	Applies supported crop parameters through a common organ interface; validity remains configuration-specific.
Architecture and object set	<code>initiation_method</code> , <code>useStaticArc</code> ; procedural, imported, reduced, static, dynamic or continued plant states	Matches an isolated organ, representative system, whole plant or replicated planting. Geometry and topology exist only for represented objects.
Environment and radiation	<code>inputEnvironmentCondition</code> , climate settings; direct, empirical or ray-traced interception through <code>calcLightInterception</code> and <code>lightInterceptionMode</code>	Supports field or controlled forcing, efficient ensembles or local exposure analysis. Forcing is a boundary condition, not a simulated microclimate.
Gas exchange and hydraulics	<code>calcPotential_PT</code> , <code>calcActual_PT</code> ; stomatal, soil and root settings	Compares potential with hydraulically constrained gas exchange at the resolution of the configured object set.
Carbon allocation or transport	<code>calcCarbonAllocation</code> , <code>useCTRAM</code> ; disabled, common-pool carbon allocation or graph-resolved carbon transport	The carbon process service uses common-pool carbon allocation for aggregate allocation or graph-resolved carbon transport for topology- and organ-resolved variability. The representations share potential demands but not regulatory state variables or solvers.
Fruit physiology	<code>fruitModule</code> : <code>simpleFruit</code> , <code>complexBerry</code> or <code>virtualFruit</code> ; former Boolean selectors are compatibility fallbacks	Selects demand-only growth, coupled berry physiology or JFruit2. Backends differ in state detail and parameter requirements.
Development and management	Phenology, organ initiation, pruning and planer controls	Holds structure fixed or tests developmental and management-induced graph change.
Outputs, horizon and cadence	<code>tables</code> , <code>useOutputTable</code> , <code>useOrganArrays</code> ; organ, diagnostic, simulation-time and output-cadence settings	Exports organ-to-field states for hourly or longer analyses. Variables depend on active modules; reporting does not change the hourly integration step.

The DAI 2012 12-leaf grapevine scenario is provided as a worked example of the four principal configuration documents: model options (Supplementary File S1), initial conditions (Supplementary File S2), plant parameters (Supplementary File S3) and soil parameters (Supplementary File S4). Values are unchanged from the executable scenario files, while comments describe parameter purpose and documented units. The files use the `.jsonc` extension because standard JSON does not permit comments; FruitCropXL simulations continue to use the corresponding comment-free `.json` files. Annotation provenance and unresolved documentation gaps are recorded in the accompanying README.

2 Additional physiological equations and implementation details

2.1 Leaf biochemical capacity, electron transport and energy balance

The main Methods section “Leaf gas exchange and leaf energy balance” retains the defining Rubisco- and electron-transport-limited net-assimilation equations together with the stomatal and hydraulic-response equations. The implementation-level inputs to those equations and the leaf-energy-balance calculations are given here.

2.1.1 Leaf nitrogen and biochemical capacity

Leaf nitrogen is an important determinant of photosynthetic biochemical capacity and can change as leaves develop and age, while leaf mass per area also responds to the local light environment. FruitCropXL therefore includes an optional dynamic leaf-nitrogen pathway that represents these two responses separately and combines them to obtain area-based leaf nitrogen, following the relationships described by Prieto et al. (2012). For this pathway,

$$N_{m,l} = \text{clip} [N_{m,0,l} + a_N(TT - TT_0), N_{m,\min}, N_{m,\max}], \quad (1)$$

$$\begin{aligned} LMA_l = \text{clip} \left[a_M \ln \{ \max(PPFD_{10,l}, 0.1) \} + b_M, \right. \\ \left. LMA_{\min}, LMA_{\max} \right], \quad (2) \\ N_{a,l} = N_{m,l} LMA_l. \end{aligned}$$

The pathway separates seasonal change in leaf nitrogen mass fraction from acclimation of leaf mass per area to the local light environment. The dynamic pathway separates seasonal change in leaf nitrogen mass fraction from acclimation of leaf mass per area to the local light environment. Starting from the reference value $N_{m,0,l}$ at thermal time TT_0 , leaf nitrogen mass fraction $N_{m,l}$ changes linearly with accumulated thermal time at the configured rate a_N , subject to lower and upper bounds. Thus, the sign and magnitude of a_N determine the direction and rate of the seasonal nitrogen response. Leaf mass per area LMA_l is calculated independently from the 10-d mean daily incident PAR, $PPFD_{10,l}$, and is likewise bounded. Their product gives area-based leaf nitrogen $N_{a,l}$, which links leaf nitrogen status and light acclimation to biochemical capacity. This formulation follows the relationships used by Prieto et al. (2012), who found that leaf nitrogen mass fraction changed with leaf ageing, whereas LMA responded strongly to local irradiance, and that area-based nitrogen was a useful predictor of within-canopy variation in photosynthetic capacity. Once the configured leaf-expansion threshold is reached, LMA_l is fixed, whereas $N_{m,l}$ can continue to respond to thermal time.

At 25 °C, area-based nitrogen determines the reference biochemical capacities, with an additional bounded NSC feedback on V_{cmax} and J_{max} :

$$V_{\text{cmax},25,l} = \eta_l(1 - r_{\text{NSC},l})s_V(N_{a,l} - N_{a,\min}), \quad (3)$$

$$J_{\text{max},25,l} = \eta_l(1 - \alpha_J r_{\text{NSC},l})s_J(N_{a,l} - N_{a,\min}), \quad (4)$$

$$R_{d,25,l} = s_R(N_{a,l} - N_{a,\min}). \quad (5)$$

$$\kappa_{2,l} = s_\kappa N_{a,l} + b_\kappa, \quad (6)$$

$$r_{\text{NSC},l} = \min(r_{\text{NSC},\text{max}}, \beta_{\text{NSC}} f_{\text{NSC},l}). \quad (7)$$

Here $N_{m,l}$, $N_{m,0,l}$, $N_{m,\min}$ and $N_{m,\text{max}}$ are leaf nitrogen mass fractions (g N g⁻¹ dry matter); TT and TT_0 are current and reference thermal time (degree-days); and a_N is the fitted nitrogen response (g N g⁻¹ dry matter degree-day⁻¹). LMA_l , $LMA_{\min}$, LMA_{max} and b_M are leaf mass per area (g dry matter m⁻²), while a_M has the same leaf-mass-per-area unit per unit of the logarithmic predictor. The executable evaluates the logarithm from the numerical value of $PPFD_{10,l}$ expressed in mol photons m⁻² d⁻¹, with 0.1 in that same numerical unit as its lower bound; this calibrated numerical convention should not be interpreted as taking the logarithm of a dimensional quantity. $N_{a,l}$ and $N_{a,\min}$ are area-based nitrogen contents (g N m⁻²).

The additional NSC term provides an empirical feedback from leaf carbohydrate status to photosynthetic capacity adapted from the QualiTree formulation (Lacroix et al. 2024). Here $f_{\text{NSC},l}$ is total leaf NSC divided by leaf dry matter; starch and soluble sugars are not distinguished in this feedback. Increasing $f_{\text{NSC},l}$ reduces V_{cmax} through $r_{\text{NSC},l}$, bounded by $r_{\text{NSC},\text{max}}$, while J_{max} responds more weakly through the relative sensitivity α_J . The apple defaults are $\beta_{\text{NSC}} = 1.81$, $r_{\text{NSC},\text{max}} = 0.60$ and $\alpha_J = 0.30$. The direction of this feedback is consistent with reported photosynthetic down-regulation under restricted sink demand in rice and with reduced photochemical performance under sink limitation in apple (Wünsche et al. 2005; Fabre et al. 2019). The relative J_{max} response is an empirical model parameterisation and should not be interpreted as a directly calibrated apple relationship or a universal constant.

The organ-type multiplier η_l , total leaf NSC fraction $f_{\text{NSC},l}$, empirical slope β_{NSC} , NSC inhibition $r_{\text{NSC},l}$, maximum inhibition $r_{\text{NSC},\text{max}}$ and relative J_{max} sensitivity α_J are dimensionless. $V_{\text{cmax},25,l}$ and $R_{d,25,l}$ are in $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$, $J_{\text{max},25,l}$ is in $\mu\text{mol electrons m}^{-2} \text{ s}^{-1}$, and $\kappa_{2,l}$ and b_κ are in mol electrons mol⁻¹ photons. Accordingly, s_V and s_R are in $\mu\text{mol CO}_2 \text{ g}^{-1} \text{ N s}^{-1}$, s_J is in $\mu\text{mol electrons g}^{-1} \text{ N s}^{-1}$, and s_κ is in mol electrons mol⁻¹ photons m² g⁻¹ N.

2.1.2 Electron-transport solution

The electron-transport rate J_l determines the light-limited supply of reducing power used in the electron-transport-limited photosynthesis calculation. For each leaf, FruitCropXL combines incident PAR with the leaf-specific maximum electron-transport capacity $J_{\text{max},l}$ and low-light conversion efficiency $\kappa_{2,l}$ defined above. Their response is represented by the non-rectangular hyperbola (Yin and Struik 2009; Evers et al. 2010):

$$\theta J_l^2 - (\kappa_{2,l} I_{\text{inc},l} + J_{\text{max},l}) J_l + \kappa_{2,l} I_{\text{inc},l} J_{\text{max},l} = 0, \quad (8)$$

The executable uses the smaller root of Equation 7. At low irradiance, J_l is governed primarily by incident PAR and $\kappa_{2,l}$, whereas at high irradiance it approaches the capacity set by $J_{\max,l}$; the dimensionless parameter θ controls the curvature of this transition. Here $I_{\text{inc},l}$ is incident PAR ($\mu\text{mol photons m}^{-2} \text{s}^{-1}$), J_l and $J_{\max,l}$ are electron-transport rates ($\mu\text{mol electrons m}^{-2} \text{s}^{-1}$), and $\kappa_{2,l}$ is in $\text{mol electrons mol}^{-1} \text{photons}$. The resulting J_l is passed to the electron-transport-limited net-assimilation calculation described in the main Methods.

2.1.3 Penman–Monteith transpiration and leaf temperature

Leaf transpiration follows the modified Penman–Monteith formulation in Equation 8:

$$E_\ell = \frac{S(T_\ell, T_a)R_{n,\ell} + \rho c_p D_a / r_{a,\ell}}{\lambda \left[\Delta(T_a) + \gamma \left(1 + \frac{r_{s,\ell}}{r_{a,\ell}} \right) \right]}, \quad r_{s,\ell} = \frac{1}{g_{s,w,\ell}^{(p)}}. \quad (8)$$

Here E_ℓ is per unit leaf area (mm s^{-1} , numerically $\text{kg water m}^{-2} \text{s}^{-1}$); $D_a = e_s(T_a) - e_a$ is atmospheric vapour-pressure deficit (kPa); r_s and r_a are stomatal and aerodynamic resistance (s m^{-1}); λ is latent heat of vaporisation; ρc_p is volumetric heat capacity of air; and γ is the psychrometric constant. The executable calculates r_a as the sum of its leaf boundary-layer and turbulence resistances from local wind, leaf width and height, while r_s is the reciprocal of physical water-vapour conductance.

With saturation vapour pressure $e_s(T)$ (kPa), the second-pass secant slope and its equal-temperature limit are

$$S(T_\ell, T_a) = \frac{e_s(T_\ell) - e_s(T_a)}{T_\ell - T_a}, \quad (9)$$

$$S(T_a, T_a) = \Delta(T_a) = \left. \frac{de_s}{dT} \right|_{T=T_a}. \quad (10)$$

The first pass uses $\Delta(T_a)$ directly; the second evaluates the secant expression without a near-equality tolerance. Net radiation is

$$R_{n,\ell} = Q_{\text{abs},\ell} + R_{\text{lw},\ell}(T_\ell, T_a), \quad (11)$$

where $Q_{\text{abs},\ell}$ is absorbed short-wave radiation and $R_{\text{lw},\ell}$ is the signed net two-sided long-wave flux calculated from the Stefan–Boltzmann difference between leaf and air temperature; all three radiative fluxes are in $\text{J m}^{-2} \text{s}^{-1}$. Leaf temperature is updated by

$$T_\ell = T_a + \frac{r_{a,\ell} [R_{n,\ell} - \lambda E_\ell]}{\rho c_p}, \quad (12)$$

where the energy-balance term gives a temperature difference in K, numerically equal to $^\circ\text{C}$. In the executable `calcTleaf` update, the radiation function is evaluated with both temperature arguments set to T_a ; the second-pass Penman–Monteith calculation separately evaluates R_n at the current T_l and T_a .

2.1.4 Two-pass update and outputs

The first pass evaluates biochemical assimilation, stomatal conductance and Penman–Monteith transpiration at $T_l = T_a$, then updates T_l . The second pass recalculates biochemical capacity and stomatal conductance at that temperature, evaluates the secant saturation slope,

and updates transpiration and leaf temperature. These potential states enter the hydraulic solver. After a hydraulic solution is accepted, FruitCropXL converts accepted stomatal conductance to physical units, recalculates transpiration and leaf temperature, reruns the biochemical calculation and applies the diffusion-resistance ratio in the main realised-assimilation equation. Principal outputs are potential and realised assimilation and water flux, stomatal conductance, leaf temperature, leaf water potential and hydraulic-loss indicators.

2.2 Additional hydraulic, soil and root equations

For soil water content θ , saturated content θ_s , retention parameters α and n , and conductivity-shape parameter p , the implemented retention and conductivity relations are

$$W = \text{clip} \left(\frac{\theta}{\theta_s}, 0, 0.999 \right), \quad (13)$$

$$\psi_s = -\frac{[1 - W^{n/(n-1)}]^{1/n}}{1000\alpha W^{1/(n-1)}}, \quad (14)$$

$$W = [1 + \{\alpha(-1000\psi_s)\}^n]^{(1-n)/n}, \quad \theta = \theta_s W, \quad (15)$$

$$M = [1 + \{\alpha(-1000\psi_s)\}^n]^{-1}, \quad (16)$$

$$K_s(\psi_s) = K_{sat} M^{p-p/n} [1 - (1 - M)^{1-1/n}]^2. \quad (17)$$

Here W is relative volumetric water content and M is the dimensionless conductivity state; θ and θ_s are current and saturated volumetric water contents (m^3 water m^{-3} soil). The retention exponent n and conductivity-shape parameter p are dimensionless. Soil water potential ψ_s is in MPa and α is in kPa^{-1} ; the factor 1000 converts MPa to kPa inside the retention relations. K_s and the internally converted K_{sat} are in $\text{m}^2 \text{MPa}^{-1} \text{s}^{-1}$.

For root compartment r , non-negative uptake weights and current soil-root resistance are

$$w_r = \max \left\{ 0, \alpha_r^* \left(G_{r,\max} + \frac{1}{R_{sp,r}} \right) (\psi_{s,r} - \psi_{rx,r}^{\text{prev}}) \right\}, \quad f_r = \frac{w_r}{\sum_k w_k}, \quad (18)$$

$$R_{sp,r} = \min \left[R_{SR} q_{R,r} \frac{K_{sat,r}}{K_{s,r}}, R_{sp,\max} \right]. \quad (19)$$

For $F_r = f_r F_w$, the trial potentials and biomass-dependent conductance are

$$\psi_{rs,r} = \psi_{s,r} - R_{sp,r} F_r, \quad (20)$$

$$\psi_{rx,r} = \psi_{rs,r} - \frac{F_r}{G_r}, \quad (21)$$

$$G_{r,\max} = a_{G,r} M_r + b_{G,r}, \quad (22)$$

$$G_{r,\text{tr}} = \text{clip}_{G_{r,\min}}^{G_{r,\max}} (\tau_{\text{tr},r} F_r), \quad (23)$$

$$G_r = \text{clip}_{G_{r,\min}}^{G_{r,\max}} \{ [G_{r,\min} + G_{r,\text{circ}} + G_{r,\text{tr}}] (1 + f_{ABA,r}) \}. \quad (24)$$

Here w_r is a non-negative potential uptake weight (mg water s^{-1}) and f_r is its dimensionless normalised fraction; the implementation guards the zero-sum case. α_r^* is the dimensionless wet-side response, $G_{r,\max}$ is maximum root conductance, $R_{sp,r}$ is soil-root resistance, and $\psi_{s,r} - \psi_{rx,r}^{\text{prev}}$ is the soil-to-previous-root-xylem potential difference. R_{SR} and $R_{sp,\max}$ are the saturated baseline and upper-bound resistances ($\text{MPa s mg}^{-1} \text{ water}$), $q_{R,r}$ is a dimensionless root-density resistance factor, and $K_{sat,r}/K_{s,r}$ is a dimensionless conductivity ratio.

$F_r = f_r F_w$ is compartment water flux (mg water s^{-1}); $\psi_{rs,r}$, $\psi_{s,r}$ and $\psi_{rx,r}$ are root-surface, soil and root-xylem water potentials (MPa). G_r , $G_{r,\max}$, $G_{r,\min}$, $G_{r,\text{tr}}$, $G_{r,\text{circ}}$ and $b_{G,r}$ are conductances ($\text{mg water s}^{-1} \text{ MPa}^{-1}$). Fine-root dry mass M_r is in mg dry matter , so $a_{G,r}$ is conductance per mg dry matter . The transpiration-response coefficient $\tau_{tr,r}$ is in MPa^{-1} , and the bounded ABA response $f_{ABA,r}$ is dimensionless. The circadian and ABA components are configuration dependent.

For each whole-plant flux candidate, the root module calculates the PlantBase xylem water potential directly as $\psi_{PB} = \sum_r f_r \psi_{rx,r}$. The accepted value is the root-shoot hydraulic boundary. Although structural roots define an axial water-potential update, the accepted hydraulic path inspected here does not call it; structural-root axial and gravitational propagation is therefore inactive in this path.

For above-ground segments, the signed gravitational head retained in the main water-potential equation is $\rho_w g \Delta z / 10^6 \simeq 0.00961 \Delta z \text{ MPa}$, using $\rho_w = 980 \text{ kg m}^{-3}$. The executable evaluates the equivalent expression $10^{-3} D_w g \Delta z$ with $D_w = 0.98 \text{ g cm}^{-3}$ and $g = 9.81 \text{ m s}^{-2}$. Annual-internode conductance uses at least the configured conducting-pipe diameter floor, normally 0.0015 m , before applying the allometry. The exponent $c_{x,3}$ is dimensionless and is typically within 2.0 – 5.0 . The empirical scale coefficient $c_{x,2}$, typically 2.5 – 2.8 , carries the implied units of the implemented metre-based allometry; its value is therefore tied to this numerical-unit convention. The factor 10^6 converts the resulting base conductance scale to $\text{mg water s}^{-1} \text{ MPa}^{-1}$ (Tyree and Zimmermann, 2002).

Leaf hydraulic conductance responds to candidate leaf water flux and xylem ABA before the PLC correction. Its raw value is

$$K_l^{\text{raw}} = \text{clip}_{G_{xl,\min}}^{G_{xl,\max}} \left\{ \left[G_{xl,0} + \text{clip}_{G_{xl,\min}}^{G_{xl,\max}} (\tau_{tr,l} F_l) \right] \left[1 + \tau_{ABA,l} \ln \left(\frac{[ABA]_x}{[ABA]_{\text{ref}}} \right) \right] \right\}. \quad (25)$$

Here K_l^{raw} , $G_{xl,0}$, $G_{xl,\min}$ and $G_{xl,\max}$ are conductances ($\text{mg water s}^{-1} \text{ MPa}^{-1}$), F_l is candidate leaf water flux (mg water s^{-1}), $\tau_{tr,l}$ has unit MPa^{-1} and $\tau_{ABA,l}$ is dimensionless. $[ABA]_x$ and $[ABA]_{\text{ref}} = 20 \mu\text{mol m}^{-3}$ have the same unit. In the active candidate path, raw conductance reads the ABA state stored on PlantBase; root-side trial ABA is passed separately to the stomatal response.

Effective conductance and candidate leaf water potential are

$$K_l = \max[K_l^{\text{raw}} (1 - PLC_l), K_{l,\min}], \quad (26)$$

$$\psi_l^{\text{trial}} = \min \left[\psi_{\text{PB}}^{\text{trial}}(F_w), \max \left(-4, \psi_{\text{parent},l}^{t-1} - \frac{F_l^{\text{pot}}(F_w/F_w^{\text{pot}})}{K_l} \right) \right], \quad (27)$$

where $K_{l,\min} = G_{xl,\min}$ is the conductance floor and the candidate water potential is bounded below at -4 MPa and above by the current-trial PlantBase boundary. Raw conductance is clipped between $G_{xl,\min}$ and $G_{xl,\max}$ before the post-PLC floor is applied. If $F_w^{\text{pot}} \leq 0$ or is non-finite, the candidate and the ratio F_w/F_w^{pot} are set to zero. Bracketing may evaluate $F_w > F_w^{\text{pot}}$, but the accepted realised flux is clamped to $0 \leq F_w \leq F_w^{\text{pot}}$ before commitment.

2.3 Units, state variables and sign conventions

FruitCropXL uses carbon mass as the common accounting currency for source supply, sink demand, reserves and transport. Accordingly, carbon terminology is used for bookkeeping amounts and fluxes, whereas sugar terminology is reserved for specific transported compounds and regulatory concentration states. Structural and reserve states are stored internally as carbon mass, whereas the regulatory variables presented to source and sink response functions are sugar concentrations. The principal quantities are:

Table S2: Carbon variables and units used in the allocation and transport methods.

Quantity	Definition	Implemented or documented unit
$C_o^{\text{str}}, C_o^{\text{res}}$	organ structural and reserve or non-structural carbon states	mg C
$S_{A,l}$	assimilation-derived carbon supplied by leaf l during the step	mg C per hourly step
$D_{o,p}^{\text{pot}}$	potential demand of process p in organ o	mg C per hourly step
L_o^{pot}, L_o	potential and concentration-regulated source loading	mg C per hourly step
$U_{o,p}$	realised unloading to process p	mg C per hourly step
Q_o^{leak}	passive leakage from the common pool into organ reserve	mg C per hourly step
C_{pool}	spatially uniform common-pool sugar concentration, represented as a regulatory sugar mass fraction	g sugar g ⁻¹ phloem sap
$c_{C,o}$	graph-local phloem carbon concentration	g C cm ⁻³
$c_{\text{sug},o}$	graph-local phloem sugar concentration after species-dependent conversion	g sugar cm ⁻³ phloem sap
Φ_{PB}	graph-resolved PlantBase carbon-potential boundary	(g C cm ⁻³) ²
$c_{\text{sug,PB}}$	PlantBase phloem sugar concentration converted from Φ_{PB}	g sugar cm ⁻³ phloem sap
ρ_{sap}	phloem-sap density used to relate sugar mass fraction and volumetric concentration	g phloem sap cm ⁻³
$\Phi_o = c_{C,o}^2/2j_o$	graph-local carbon potential signed graph-resolved organ exchange	(g C cm ⁻³) ² g C h ⁻¹
a_o, b_o	intercept and slope of the local j_o - Φ_o linearisation	g C h ⁻¹ ; g C h ⁻¹ per unit carbon potential
A, B, A', B'	locally aggregated and upstream-transformed transport coefficients	same respective units as a_o and b_o

Common-pool source and sink terms are accumulated as carbon amounts over the hourly model step, whereas graph-resolved carbon transport is assembled as a carbon rate. The

graph-resolved implementation therefore converts step-based carbon amounts from mg C per step to the g C h⁻¹ units required by the transport solver. Common-pool loading, leakage and unloading are stored as separate non-negative terms. In graph-resolved carbon transport, $j_o > 0$ denotes unloading from the phloem to a sink and $j_o < 0$ denotes loading into the phloem by a source.

For graph-resolved carbon transport, the implementation converts apple carbon concentration to the configured transported-sugar mixture of 75% sorbitol and 25% sucrose; this proportion is a modelling assumption rather than a fitted or universally fixed biological value. Grapevine uses sucrose alone. Consequently, C_{pool} and $c_{\text{sug},o}$ have different concentration bases: the former is sugar mass per solution mass and the latter sugar mass per sap volume. The dimensional relationship and configured unit-density convention are defined in the main Methods under “Graph-resolved carbon transport”, but the implementation does not convert reported values between configurations. The current implementation directly reuses the same configured numerical K_m values without a separate density conversion; each K_m therefore inherits the concentration basis active in the selected representation. Common-pool and graph-resolved concentration outputs retain their own definitions and units and are not numerically interchangeable.

2.4 Assimilation supply, maintenance demand, loading and unloading

Realised area-based net assimilation is converted to carbon supplied by leaf l during the hourly step using Equation 28:

$$S_{A,l} = 10^{-3} A_{n,l}^{\text{real}} A_l \Delta t_s M_C, \quad S_A^{\text{plant}} = \sum_{l=1}^{n_l} S_{A,l}, \quad (28)$$

where $S_{A,l}$ is the carbon amount supplied by leaf l (mg C per hourly step), $A_{n,l}^{\text{real}}$ is accepted realised net assimilation ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$), A_l is leaf area (m^2), $\Delta t_s = 3600 \text{ s}$ is the step duration, $M_C = 12 \mu\text{g C } \mu\text{mol}^{-1} \text{ CO}_2$ is the molar carbon conversion, and 10^{-3} converts $\mu\text{g C}$ to mg C. Thus assimilation multiplied by area and time gives $\mu\text{mol CO}_2 \text{ step}^{-1}$, multiplication by M_C gives $\mu\text{g C step}^{-1}$, and the final factor gives mg C step⁻¹. S_A^{plant} is the summed plant assimilation supply (mg C per hourly step), n_l is the number of leaves and l is the leaf index. Positive realised assimilation contributes potential leaf source loading; reserve hydrolysis provides a second potential source in leaves, perennial axes and roots where enabled.

For a generic growing organ o , potential maintenance demand before the NSC response is given by Equation 29:

$$R_{m,o}^{\text{pot}} = C_o^{\text{str}} q_{m,o} Q_{10,m}^{(T_a - T_{\text{ref}})/10}, \quad (29)$$

and the implemented availability adjustment follows Equation 30:

$$\begin{aligned} R_{m,o} &= R_{m,o}^{\text{pot}} f_{C,o}, \\ f_{C,o} &= m_{\min} + \min \left[1 - m_{\min}, \max \left(0, \frac{C_o^{\text{res}}}{C_o^{\text{str}}} \right) \right], \\ m_{\min} &= 0.25. \end{aligned} \quad (30)$$

Here $R_{m,o}^{\text{pot}}$ and $R_{m,o}$ are potential and realised maintenance carbon amounts (mg C per hourly step), C_o^{str} is structural carbon (mg C), and $q_{m,o}$ is the step-specific maintenance coefficient (h^{-1} , equivalently per one-hour step in this implementation). $Q_{10,m}$ is a dimensionless temperature coefficient; T_a and T_{ref} are air and reference temperature ($^{\circ}\text{C}$), so $(T_a - T_{\text{ref}})/10$ and the exponentiated multiplier are dimensionless. The reserve fraction $C_o^{\text{res}}/C_o^{\text{str}}$, availability response $f_{C,o}$ and floor $m_{\text{min}} = 0.25$ are dimensionless. Thus $f_{C,o}$ rises linearly from 0.25 and saturates at one. The lower fraction is a conservative modelling approximation: carbohydrate availability and respiration can be approximately related, while assimilate-starvation studies reported residual maintenance respiration of approximately 28–34% of normal night respiration (Noguchi 2005; Peraudeau et al. 2015). The response shape and saturation point were not fitted to those studies. Leaves are specialised: during light periods, negative realised assimilation is treated as maintenance and non-negative assimilation gives zero separate maintenance; at night they use the generic expression. The general maintenance formulation follows the perennial-carbon model lineage (Zhu et al. 2021a).

Potential sink demand in both carbon representations is regulated by the Michaelis–Menten response defined in the main Methods under “Source and sink regulation”. The regulatory concentration is $x_o = C_{\text{pool}}$ in common-pool carbon allocation and $x_o = c_{\text{sug},o}$ in graph-resolved carbon transport. Common-pool allocation applies this response to step-based potential demands and realised unloading amounts, whereas graph-resolved transport converts the corresponding exchanges to transport-rate units during assembly. $D_{o,p}^{\text{pot}}$ is potential demand, $U_{o,p}$ is realised unloading, and $K_{m,o,p}$ is the half-response concentration on the concentration basis used by the selected representation.

In graph-resolved transport, the nonlinear sink response is locally linearised during assembly of the transport equations. Numerical safeguards are applied during this calculation to prevent singular or degenerate coefficients, but these safeguards are solver-specific and do not modify the biological sink response. After graph convergence, realised unloading is reevaluated using the original Michaelis–Menten function. Separate K_m values are used for primary growth, secondary growth, maintenance and reserve synthesis, while fruit backends and fine-root exudation provide specialised potential demands.

Source loading follows the decreasing logistic response defined in the main Methods under “Source and sink regulation”. The same biological concentration dependence is used in common-pool carbon allocation and graph-resolved carbon transport, with $x_o = C_{\text{pool}}$ and $x_o = c_{\text{sug},o}$, respectively. L_o^{pot} and L_o are the potential and realised source-loading magnitudes, $C_{0,o}$ is the half-response concentration on the same concentration basis as x_o , and β_o determines the response steepness. In common-pool allocation, L_o is a non-negative step-based loading amount. In graph-resolved transport, the corresponding exchange is converted to transport-rate units and enters the transport solution with the sign convention $j_{\text{source},o} = -L_o$; sink unloading is positive. Numerical bounds used during evaluation and linearisation are implementation safeguards and do not alter the underlying biological loading function.

The logistic response determines how increasing sugar concentration restricts further source loading. Reserve-derived loading is additionally subject to organ-specific eligibility rules based on the configured minimum reserve fraction. Generic growing organs and leaves constrain export so that reserve status does not fall below this minimum, whereas internodes, cordons, trunks and structural roots use the pre-step reserve criterion. Fine roots use the same pre-

step criterion together with a numerical fallback when reserve-derived loading is unavailable. These source–sink responses follow the formulation used in the earlier comparison of common-pool allocation and graph-resolved phloem transport (Zhu et al. 2021a).

The whole-plant loading-minus-leakage-minus-unloading balance used by common-pool carbon allocation is defined in the main Methods under “Common-pool carbon allocation”. The carbon process service searches for the regulatory concentration C_{pool} at which this residual is zero. All terms are compatible carbon amounts accumulated over the hourly model step: source loading supplies the balance, whereas passive leakage into organ reserves and sink unloading remove carbon from it. C_{pool} is therefore a regulatory sugar concentration rather than a stored carbon stock. If an accepted numerical root cannot be obtained, selection of a bounded finite fallback remains a solver safeguard rather than an alternative biological allocation rule.

2.5 Reserve synthesis, storage and remobilisation

Leaves, annual internodes, cordons, trunks, fine roots and structural roots contain an organ NSC or reserve-carbon pool C_o^{res} (mg C). For generic non-leaf storage organs, maximum reserve carbon is defined by Equation 31:

$$C_{o,\text{max}}^{\text{res}} = M_o f_{\text{max},o} f_{C,\text{suc}}, \quad (31)$$

where $C_{o,\text{max}}^{\text{res}}$ is reserve capacity (mg C), M_o is organ dry mass (mg dry matter), and $f_{\text{max},o}$ and the sucrose carbon fraction $f_{C,\text{suc}}$ are dimensionless. Capacity-driven potential synthesis follows Equation 32:

$$S_o^{\text{pot}} = \max\{0, k_{\text{syn},o} (C_{o,\text{max}}^{\text{res}} - C_o^{\text{res}}) f_{\text{enz}}(T_a)\}, \quad (32)$$

where S_o^{pot} is potential reserve-synthesis demand (mg C per hourly step), $k_{\text{syn},o}$ is the synthesis rate (h^{-1}) and $f_{\text{enz}}(T_a)$ is a dimensionless enzyme-temperature response. Reserve synthesis is treated as a sink process, and realised storage is obtained by applying the main-text Michaelis–Menten sink response to potential synthesis demand, using the reserve-specific half-response concentration $K_{m,o,\text{res}}(d)$:

$$S_o = S_o^{\text{pot}} f_{\text{sink},o,\text{res}}(x_o), \quad (33)$$

where S_o is realised reserve storage (mg C per hourly step), $x_o = C_{\text{pool}}$ in common-pool carbon allocation and $x_o = c_{\text{sug},o}$ in graph-resolved carbon transport. $K_{m,o,\text{res}}$ has the concentration unit and basis of the active representation. During transport assembly, the reserve-synthesis sink term is linearised with the numerical 10^{-6} activity floor described in Section 2.4; final realised storage is reevaluated with the floor-free main-text response. Potential synthesis therefore decreases as the reserve pool approaches its organ-specific capacity and is constrained to be non-negative. When the current pool equals or exceeds capacity, potential synthesis is zero. Capacity-limited storage and the separate representation of reserve synthesis, remobilisation and loading follow earlier functional–structural and perennial-crop formulations (Cieslak et al. 2011; Zhu et al. 2021b, 2021a).

Seasonal reserve status is represented by the normalised beta growth function in Equation 34:

$$\beta_{\text{res}}(d) = \begin{cases} 0, & d \leq 0, \\ \text{clip} \left[\left(1 + \frac{t_{e,r}-d}{t_{e,r}-t_{m,r}} \right) \left(\frac{d}{t_{e,r}} \right)^{t_{e,r}/(t_{e,r}-t_{m,r})}, 0, 1 \right], & 0 < d \leq t_{e,r}, \\ 1, & d > t_{e,r}, \end{cases} \quad (34)$$

where d is elapsed time from the seasonal origin used by the environment clock, implemented in days since winter solstice; $t_{m,r}$ and $t_{e,r}$ are the times (days) at which the seasonal-state rate is maximal and the transition is complete, respectively. Ratios of these time quantities and their exponents are dimensionless, and $\beta_{\text{res}}(d)$ is a dimensionless interpolation state bounded between zero and one; here subscript r denotes the reserve-season parameterisation, not a root compartment. The beta function is an empirical seasonal interpolation kernel rather than a mechanistic model of starch biochemistry. A related normalised beta function was previously used to represent seasonal variation in the NSC synthesis-rate coefficient k_{syn} of the APSIM grapevine model (Zhu et al. 2021b). In FruitCropXL, the resulting state instead interpolates the implemented reserve-storage affinity `km_res` and effective hydrolysis coefficient through Equation 35 and Equation 36:

The active reserve-storage affinity is interpolated between configured early- and late-season values:

$$K_{m,o,\text{res}}(d) = K_{m,o}^{\text{early}} + (K_{m,o}^{\text{late}} - K_{m,o}^{\text{early}})\beta_{\text{res}}(d). \quad (35)$$

The reserve-hydrolysis coefficient is interpolated independently between its early- and late-season values:

$$k_{\text{hyd},o}(d) = k_{\text{hyd},o}^{\text{early}} + (k_{\text{hyd},o}^{\text{late}} - k_{\text{hyd},o}^{\text{early}})\beta_{\text{res}}(d). \quad (36)$$

Here $K_{m,o}^{\text{early}}$ and $K_{m,o}^{\text{late}}$ are the configured early- and late-season reserve-synthesis half-response concentrations, and $k_{\text{hyd},o}^{\text{early}}$ and $k_{\text{hyd},o}^{\text{late}}$ are the corresponding reserve-hydrolysis coefficients (h^{-1}). Thus, the same seasonal state coordinates increasing reserve-storage priority with declining reserve remobilisation, while the reserve-synthesis coefficient $k_{\text{syn},o}$ remains independent of this interpolation.

Potential reserve hydrolysis and concentration-regulated source loading are then

$$H_o^{\text{pot}} = C_o^{\text{res}} k_{\text{hyd},o}(d) f_{\text{enz}}(T_a), \quad L_o = H_o^{\text{pot}} f_{\text{source},o}(x_o). \quad (37)$$

In the implementation, reserve-season progression coordinates both storage affinity and reserve remobilisation. The reserve-synthesis affinity $K_{m,\text{res}}$ is interpolated between the configured early- and late-season values, $K_{m,\text{res}}^{\text{early}}$ and $K_{m,\text{res}}^{\text{late}}$. The effective hydrolysis coefficient is likewise interpolated between the configured early- and late-season hydrolysis rates, $k_{\text{hyd}}^{\text{early}}$ and $k_{\text{hyd}}^{\text{late}}$. Reserve synthesis is controlled independently through the synthesis-rate coefficient k_{syn} .

This seasonal switching provides an empirical representation of the transition from reserve remobilisation towards reserve replenishment and is consistent with observed seasonal patterns of carbohydrate depletion and recovery in grapevine, although the timing and magnitude of these responses depend on crop load, canopy function and environmental conditions (Smith and Holzapfel 2009; Rossouw et al. 2017). Potential hydrolysis remains proportional

to the available reserve pool and is calculated independently of synthesis. The configured minimum NSC fraction suppresses reserve-derived export once reserves reach the lower bound; organ-specific post-loading caps, pre-step eligibility gates and the fine-root numerical fallback are described in Section 2.4. Reserve synthesis, hydrolysis, storage affinity, capacity and minimum-reserve parameters are organ- and scenario-specific rather than universal constants and are therefore retained in the configuration files.

Reserve storage, remobilisation, maintenance and growth are not advanced as an independent continuous reserve ODE. During the carbon solve, synthesis is a sink and hydrolysis is a source. A representative non-fruit reserve commit is given by Equation 38:

$$\Delta C_o^{\text{res}} = S_{A,o} + U_{o,m} + U_{o,res} + Q_o^{\text{leak}} - L_o - q_{\text{trans},o} L_o - R_{m,o}^{\text{pot}}, \quad (38)$$

where ΔC_o^{res} , $S_{A,o}$ (zero outside leaves), $U_{o,m}$, $U_{o,res}$, Q_o^{leak} , L_o and $R_{m,o}^{\text{pot}}$ are carbon amounts over the hourly step (mg C per step). The transport-cost fraction $q_{\text{trans},o}$ is dimensionless, so $q_{\text{trans},o} L_o$ has the same unit. Organ overrides omit terms that do not apply. The difference $U_{o,m} - R_{m,o}^{\text{pot}}$ makes reserve carbon cover unmet maintenance in this bookkeeping, but it does not create a maintenance-first allocation cascade. Reserve affinity is refreshed during the post-solve update and therefore affects the next hourly carbon solve.

Leaves are specialised. Their daytime loading combines realised assimilation with available reserve according to light and reserve state; nighttime hydrolysis is recalculated from computed night length to target 80% reserve degradation over the night. Their generic synthesis rate is normally zero. Principal outputs are organ and whole-plant reserve carbon, reserve fraction, seasonal β_{res} , effective K_m and hydrolysis rate, potential and realised synthesis, loading and net reserve change.

2.6 Graph-resolved carbon-transport solution

2.6.1 Local carbon potential and linearised exchange

Graph-resolved carbon transport adapts the fold-unfold method of Seleznyova and Hanan (2018) and the earlier GrapevineXL formulation (Zhu et al. 2021a). It uses the carbon potential defined in the main Methods under “Graph-resolved carbon transport”. The additional implementation-level transformation required here converts graph-local carbon concentration to transported-sugar concentration:

$$c_{\text{sug},o} = \kappa_s c_{C,o}, \quad (39)$$

Here $c_{C,o}$ is graph-local carbon concentration (g C cm^{-3}), $c_{\text{sug},o}$ is graph-local sugar concentration (g sugar cm^{-3}), and the species conversion κ_s is g sugar per g C and is dimensionless as a mass ratio.

The conversion factor is

$$\kappa_s = \begin{cases} 1/f_{C,\text{suc}}, & \text{grapevine,} \\ (1 - f_{\text{sor}})/f_{C,\text{suc}} + f_{\text{sor}}/f_{C,\text{sor}}, & \text{apple,} \end{cases} \quad (40)$$

where $f_{C,\text{suc}}$ and $f_{C,\text{sor}}$ are the carbon mass fractions of sucrose and sorbitol. The configured apple transported-sugar mixture assumes $f_{\text{sor}} = 0.75$ as a modelling choice. Local signed organ exchange is linearised around the current carbon-potential state using the first-order

Taylor approximation defined in the main Methods under “Graph-resolved carbon transport”. The implementation-specific coefficients are

$$a_o = j_o(\Phi_o^0) - \Phi_o^0 j'_o(\Phi_o^0), \quad b_o = j'_o(\Phi_o^0). \quad (41)$$

Here superscript 0 denotes the current linearisation point. j_o is signed organ exchange (g C h^{-1}), with $j_o > 0$ denoting sink unloading from the phloem and $j_o < 0$ source loading into it; $j'_o = dj_o/d\Phi_o$ is the local slope (g C h^{-1} per unit carbon potential). Hence a_o is in g C h^{-1} and $b_o = j'_o$ in g C h^{-1} per unit carbon potential. The derivative is evaluated through sugar concentration:

$$\frac{dj_o}{d\Phi_o} = \frac{dj_o}{dc_{\text{sug},o}} \frac{dc_{\text{sug},o}}{d\Phi_o}, \quad \frac{dc_{\text{sug},o}}{d\Phi_o} = \frac{c_{\text{sug},o}}{2\Phi_o}. \quad (42)$$

The first factor is g C h^{-1} per (g sugar cm^{-3}), and the second is (g sugar cm^{-3}) per unit carbon potential, so their product has the required unit of carbon rate per carbon potential.

2.6.2 Recursive folding and the guarded PlantBase carbon-potential boundary

Local coefficients and the already folded coefficients of child-side segments q are combined towards the trunk and basal root:

$$A = a_o + \sum_q A'_q, \quad B = b_o + \sum_q B'_q. \quad (43)$$

Thus A is the locally aggregated intercept (g C h^{-1}) and B is the locally aggregated slope (g C h^{-1} per unit carbon potential); a prime denotes a coefficient transformed to the upstream end of a segment.

For a segment connecting this aggregate to its upstream parent,

$$j = A + B\Phi_t, \quad \Phi_t = \Phi_b - rj - x_p, \quad (44)$$

where Φ_b and Φ_t are upstream and downstream carbon potentials, j is segment carbon flux (g C h^{-1}), r is segment transport resistance (carbon potential per g C h^{-1}), and $x_p \geq 0$ is the implemented non-negative xylem-coupling decrement (carbon-potential units). Every term in the potential-drop equation therefore has carbon-potential units. The coefficients viewed from the upstream end are

$$A' = \frac{A - Bx_p}{1 + Br}, \quad B' = \frac{B}{1 + Br}, \quad j = A' + B'\Phi_b. \quad (45)$$

A' and B' have the same respective units as A and B ; the products Br in the denominators are dimensionless, and Bx_p has units of g C h^{-1} .

The distal shoot and root relations are folded independently and meet at PlantBase:

$$A_{\text{PB}} = A'_{\text{root}} + A'_{\text{shoot}}, \quad B_{\text{PB}} = B'_{\text{root}} + B'_{\text{shoot}}. \quad (46)$$

PB denotes PlantBase. A'_{root} and A'_{shoot} are the upstream-transformed root- and shoot-side intercepts (g C h^{-1}), and B'_{root} and B'_{shoot} are the corresponding slopes (g C h^{-1} per unit carbon potential). Their sums A_{PB} and B_{PB} retain those respective units.

The raw PlantBase carbon-potential candidate is obtained from the zero-net-flow boundary defined in the main Methods under “Graph-resolved carbon transport”. Its numerator has the unit g C h^{-1} and its denominator g C h^{-1} per unit carbon potential. Before division, the implementation verifies that both coefficients are finite and that $|B_{\text{PB}}|$ exceeds the defined near-zero threshold (10^{-12} in the coefficient’s native unit). A finite raw candidate at least equal to the positive potential floor (10^{-8} in carbon-potential units) is retained only when its sugar transformation is also finite; otherwise, the most recent valid PlantBase carbon potential or the positive floor supplies the fallback. The installed Φ_{PB} is adaptively under-relaxed from the preceding iterate towards the raw candidate. The accepted concentration-scale state is $c_{\text{sug,PB}} = \kappa_s \sqrt{2\Phi_{\text{PB}}}$. The boundary residual $A_{\text{PB}} + B_{\text{PB}} \Phi_{\text{PB}}$ (g C h^{-1}) and the dimensionless Boolean fallback flag remain diagnostic outputs.

2.6.3 Unfolding, iteration and state commitment

The current PlantBase carbon potential is propagated outward through the shoot and structural-root graphs. For each segment, the stored folded coefficients reconstruct the downstream potential:

$$\Phi_t = \Phi_b(1 - rB') - rA' - x_p. \quad (47)$$

Here a prime identifies coefficients transformed to the upstream end of the segment, and each subtracted term is a carbon-potential decrement. The reconstructed local potential is converted back to carbon and sugar concentration, signed organ exchange is recalculated, and the nonlinear source and sink responses, linear coefficients, resistances and coupling terms are refreshed before the next fold–unfold iteration.

Convergence of the nonlinear iteration is quantified from the change in organ carbon flux between successive iterations:

$$\varepsilon_C = \sum_o |j_o^{(k)} - j_o^{(k-1)}|, \quad (48)$$

where k is the transport-iteration index and ε_C is the summed absolute change in organ carbon rates (g C h^{-1}). This statistic measures stabilisation of the iterative fold–unfold solution but does not by itself establish carbon mass balance.

Carbon mass-flux balance is therefore evaluated separately from the signed organ exchanges:

$$R_{\text{bal}} = \sum_o j_o, \quad \varepsilon_{\text{bal}} = |R_{\text{bal}}|, \quad (49)$$

where $j_o > 0$ denotes unloading from the phloem to a sink and $j_o < 0$ denotes loading into the phloem by a source. R_{bal} is the net carbon exchange of the graph and ε_{bal} is its absolute imbalance, both in g C h^{-1} . A balanced transport solution therefore approaches zero net source–sink imbalance.

Folding and unfolding continue until both the iterative convergence criterion, $\varepsilon_C \leq 10^{-5} \text{ g C h}^{-1}$, and the configured carbon mass-balance criterion for ε_{bal} are satisfied, or until the repeat limit is reached. These criteria provide complementary checks: ε_C tests whether organ fluxes have stabilised between iterations, whereas ε_{bal} tests whether total carbon loading and unloading are balanced in the resulting network state.

Consistency of the PlantBase solution with the folded zero-net-flow boundary is additionally monitored through

$$R_{PB} = A_{PB} + B_{PB} \Phi_{PB}, \quad (50)$$

where R_{PB} is in g C h^{-1} . The PlantBase residual therefore diagnoses the local folded boundary solution, whereas ε_{bal} evaluates mass-flux balance across the reconstructed organ exchanges.

Only after the iterative solution is accepted are local phloem sugar concentrations and organ exchanges passed to the single hourly state-commit update; trial linearisation states are not committed. If the repeat limit is reached, the final convergence statistic, carbon mass-flux imbalance and PlantBase boundary residual are retained together with the corresponding flux state for diagnostic evaluation. Reported diagnostics include iteration count, ε_C , ε_{bal} , R_{PB} , boundary-fallback status and summary concentration and flux states.

3 Software structure, repository, verification and documentation

3.1 Repository and software structure

FruitCropXL is maintained as a versioned GroIMP project. Editable XL/RGG and Java sources are under `Scripts/`; configurations and inputs are under `Model_scenarios/` and `Model_input/`; portable checks are under `tests/validation/`; and documentation and call graphs are under `Model_documents/`. The generated `Scripts/Scripts.gsz` archive is the packaged execution entry point. Reproducibility requires the source revision, archive version, runtime or container identity, scenario inputs and downstream analysis code.

GitHub Actions and corresponding local scripts execute model configurations, regression comparisons, packaging and documentation workflows. Doxygen (Van Heesch 2026) and Graphviz generate searchable XL/RGG API, inheritance and call-graph pages for local inspection and online deployment. Generated documentation supports source interpretation but is not evidence of numerical or biological validity.

3.2 Functional and packaging checks

The functional workflow advances 12 named model-option configurations for seven steps and screens compilation, configuration loading, initiation, advancement, exceptions and stalls. It also runs the embedded JUnit suite, executes the packaged archive and builds the Debian package. The JUnit inventory includes focused checks of the graph-resolved carbon-transport PlantBase boundary: finite and near-zero denominators, non-finite coefficients, fallback order, finite carbon potential and transformed PlantBase phloem sugar concentration, and normal-path equivalence. The 12-case matrix includes `model.options.FOPS-D34.json`, which activates graph-resolved carbon transport. These checks test numerical and software behaviour, not biological validity.

3.3 Regression tests and observation comparisons

The performance workflow runs six scenarios: `default`, `antony-2010`, `dai-2012-12L`, `dai-2012-3L`, `FOPS-satDryTest` and `stormTest`. Scenario YAML files select the configuration, run length, comparison variables and thresholds. The `default`, `stormTest` and FOPS cases compare current output with archived model output to detect drift. Antony additionally

compares leaf water potential and plant water flux with observations, while the DAI cases compare mean berry fresh mass, dry mass and sugar concentration. These comparisons are limited to the named variables and configurations.

Table S3: Evidence categories used to distinguish FruitCropXL software verification, regression testing, observation-based evaluation and capability demonstrations.

Evidence category	Scope	Check or comparison	Evidence sources	Interpretation and limitation
Functional and packaging verification	Twelve seven-step configurations, embedded JUnit tests, packaged <code>Scripts.gsz</code> and package build	Run completion; finite selected states; organ and root-soil invariants; graph-resolved carbon-transport PlantBase guard, finite transformed state and normal-path equivalence; archive execution and package construction	Workflow definitions, test source, logs, reports and artefacts	Verifies selected numerical boundaries, execution and packaging; does not establish seasonal or biological accuracy
Archived-output regression testing	Six named performance scenarios	Relative mean discrepancy between current and archived outputs	Scenario YAML files, thresholds, reference outputs, logs and JUnit reports	Detects implementation drift; archived outputs are not observations and changes require explicit review
Observation-based evaluation	Antony 2010 and DAI 2012 grapevine configurations	RMSE for leaf water potential and plant water flux, or mean fruit fresh mass, dry mass and sugar concentration	Observational CSV files, scenario definitions, archived logs and test reports	Limited evaluation of named variables; does not provide comprehensive or necessarily independent validation
Capability demonstrations	Compound dry-to-wet event, controlled-environment grapevine and apple crop-load scenarios presented in this paper	Process propagation, conditional factor effects and deterministic multi-scale outputs	Versioned configurations, archived outputs and analysis scripts	Demonstrates framework capabilities within defined configurations; not biological validation

Relative mean discrepancy is used for archived-output comparisons and RMSE for the named observation comparisons.

Reference-output changes require explicit scientific review; outputs were not updated for the PlantBase guard. The comparison workflow is regression evidence rather than proof of mass conservation, solver convergence or general predictive accuracy.

3.4 Automated API documentation

A separate workflow preprocesses XL/RGG module declarations for Doxygen, generates API and graph documentation and publishes the HTML site. It improves traceability but is not a strict test gate, and its tool versions are not pinned.

4 Demonstration-scenario details

4.1 Compound dry-to-wet event

The complete hourly environmental forcing used for the compound dry-to-wet event is shown in Figure S1. The main-text compound dry-to-wet response figure retains incoming radiation and prescribed soil water potential alongside the model responses, whereas the supplementary figure reports all imposed atmospheric and soil variables.

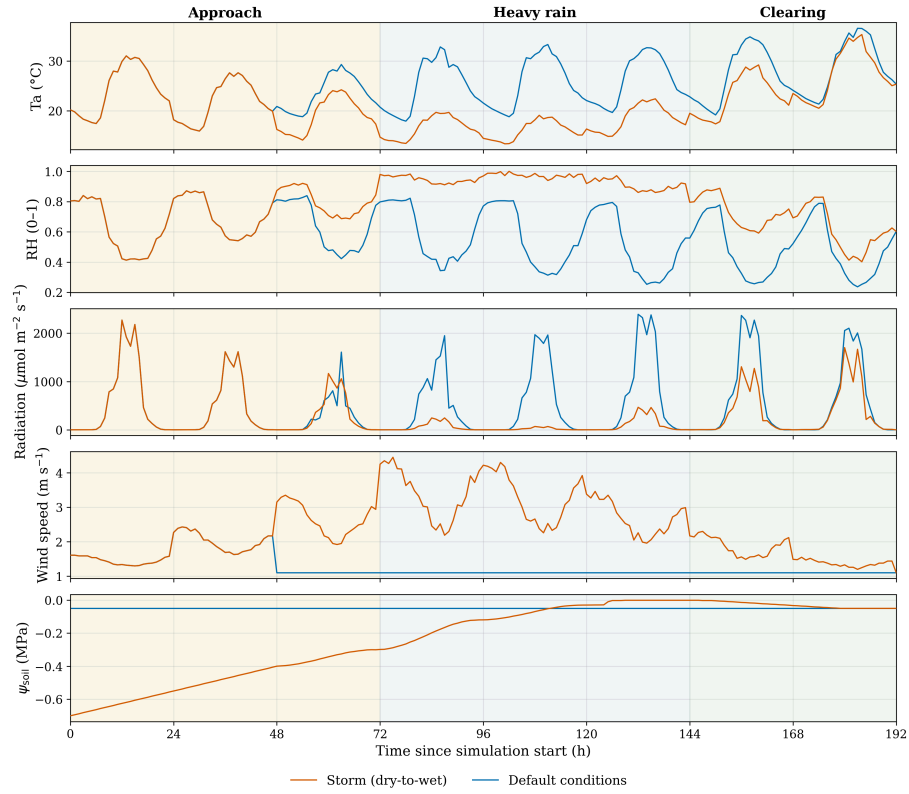


Figure S1: Complete hourly environmental forcing for the compound dry-to-wet event (orange) and default Bordeaux-based scenario (blue). Panels from top to bottom show air temperature, relative humidity, incoming total radiation, wind speed and prescribed soil water potential, ψ_{soil} . Background shading identifies the 72-h approach, 72-h heavy-rain and 48-h clearing phases. The imposed soil boundary crossed h_2 , the threshold at which the empirical wet-side reduction in root water uptake begins, and was held at h_1 , where the uptake multiplier is zero, from hours 129 to 146.

4.2 Controlled-environment grapevine scenario settings

The controlled-environment simulation experiment compared single-shoot spur and multi-shoot cane grapevines across a full-factorial combination of architecture, source size, crop load and environmental treatments. The detailed settings below complement the experimental rationale and analysis described in the main Methods section “Demonstration scenarios”.

4.2.1 Vine architecture and crop-load treatments

Table S4: Controlled-environment vine architecture and crop-load treatments.

Setting	Single-shoot/spur	configuration	Cane configuration
Shoots per vine		1	8
Bunches per shoot		1	1
Berries per bunch	50, 70, 90, 110		20, 30, 40, 50
Total berries per vine	50, 70, 90, 110		160, 240, 320, 400
Final leaves per shoot	3, 6, 9, 12		3, 6, 9, 12
Total final leaves per vine	3, 6, 9, 12		24, 48, 72, 96
In-row plant spacing		0.3 m	0.6 m
Between-row spacing		0.3 m	0.6 m
Reference leaf area	$\approx 0.25 \text{ m}^2$ at 12 leaves		$\approx 0.92 \text{ m}^2$ at 6 leaves shoot ⁻¹
Reference leaf-area-to-crop-load ratio	$0.0050 \text{ m}^2 \text{ berry}^{-1}$ at 50 berries		$0.00575 \text{ m}^2 \text{ berry}^{-1}$ at 160 berries

4.2.2 Full-factorial treatments

Table S5: Environmental and morphological levels in the core controlled-environment full-factorial simulations.

Factor	Configuration key	Core factorial levels
Daily maximum temperature	<code>Tmax</code>	25 and 30 °C
Daily minimum temperature	<code>Tmin</code>	15 and 20 °C
Atmospheric CO ₂ concentration	<code>cca</code>	400, 650, 900, and 1200 ppm
Incident daytime PAR	<code>inputPAR</code>	400, 600, 1000, and 1200 $\mu\text{mol m}^{-2} \text{ s}^{-1}$
Final leaf number	<code>ENDING_LEAF_NUMBER</code>	3, 6, 9, and 12 leaves shoot ⁻¹
Crop-load treatments	–	Four per vine configuration
Core combinations	–	256 per crop-load scenario; 2,048 across the eight crop-load scenarios

4.2.3 Fixed environmental and runtime settings

Table S6: Fixed environmental and runtime settings used in the controlled-environment simulations.

Setting	Value	Description
Environmental forcing	Internally generated	Simplified hourly greenhouse conditions rather than an external climate file
Simulation timestep	1 h	Hourly environmental and physiological updates
Simulation start	Day 35 of 2024	Initial hour: 00:00
Simulation duration	36 days	Set by <code>halt = 36</code>
Nominal photoperiod	14 h day ⁻¹	Radiation supplied during daytime and set to zero at night
PAR fraction of total radiation	0.8	Set by <code>inputfPAR</code>
Soil water potential	−0.05 MPa	Fixed well-watered condition
Wind speed	1 m s ⁻¹	Representative of controlled-environment ventilation
Maximum relative humidity	85%	Set by <code>RHmax = 0.85</code>
Minimum relative humidity	60%	Set by <code>RHmin = 0.60</code>
Daytime radiation	Selected <code>inputPAR</code> level	Constant during the nominal daytime period
Night-time radiation	0 μmol m ⁻² s ⁻¹	No incident PAR at night
Temperature maximum time	14:00	Defined by the hourly sinusoidal temperature function
Site coordinates	39.657° S, 176.879° E	Elevation: 21 m

Hourly controlled-environment air temperature was calculated with Equation 51:

$$T(h) = \frac{1}{2} \left[(T_{\max} + T_{\min}) + (T_{\max} - T_{\min}) \cos \left(\frac{\pi(h - 14)}{12} \right) \right], \quad (51)$$

where $T(h)$, $T_{\max}$ and $T_{\min}$ are air temperatures (°C) and h is hour of day (h). The cosine is evaluated from the numerical hour, so $(h - 14)/12$ represents a dimensionless phase relative to a 12-h half-period. The imposed temperature cycle reaches $T_{\max}$ at 14:00 and $T_{\min}$ at 02:00.

Table notes. The scenario files specify 50, 70, 90, and 110 berries for the four single-shoot/spur crop loads. For the cane configuration, total berry numbers are calculated from eight shoots, one bunch per shoot, and 20, 30, 40, or 50 berries per bunch. The six-leaf cane case is a factorial reference treatment; the current cane base configuration specifies nine ending leaves per shoot.

For each architecture, Type II analysis of variance used the model in Equation 52:

$$y \sim (C_{\text{CO}_2} + N_{\text{fruit}} + A_{\text{leaf}} + \bar{T}_a + R_{\text{in}})^2, \quad (52)$$

where y is the analysed response in its reported unit; C_{CO_2} is atmospheric CO_2 concentration (ppm), N_{fruit} is crop load (fruit plant⁻¹), A_{leaf} is leaf area (m² plant⁻¹), $\bar{T}_a$ is mean air temperature (°C), and R_{in} is incident PAR ($\mu\text{mol photons m}^{-2} \text{ s}^{-1}$). The squared formula notation includes all five main effects and all pairwise interactions. The partial effect size for each term was calculated with Equation 53:

$$\eta_p^2 = \frac{F df}{F df + df_{\text{res}}}, \quad (53)$$

Here F is the ANOVA statistic for the model term, df and df_{res} are the term and residual degrees of freedom, respectively, and partial η_p^2 is a dimensionless ANOVA effect-size measure. It quantifies the magnitude of that term relative to variation associated with the term plus residual variation, conditional on the fitted model. Higher values therefore indicate stronger relative effects within the sampled factorial scenario space. Partial η_p^2 values are non-additive and depend on the fitted model and sampled factor ranges; they do not sum to unity, partition total output variance or represent percentages of biological population variance. They were therefore used to compare the relative effects of factors and interactions within the deterministic simulations rather than as global sensitivity indices.

Harvest index, stored internally as `harvest_index`, was calculated from fruit biomass divided by total simulated plant biomass at the final retained record. Mean fruit dry mass and fruit sugar concentration were also taken from the final record. Fruit unloading fraction was averaged across retained records, and whole-plant biomass response was the percentage change from first to final record. Main-text leaf-number curves report the mean $\pm$ 1 SD across retained crop-load, CO_2 , radiation and temperature combinations within each architecture and leaf-number level. These bands describe dispersion across deterministic scenarios, not biological replication, parameter uncertainty or uncertainty in the mean.

Robustness to incomplete cells was checked by sorting UUIDs and retaining the same number of records in every leaf-number $\times$ crop-load cell (47 per cell for spur and 52 for cane), then refitting the same Type II models. The leading harvest-index values changed from 0.939, 0.871 and 0.191 to 0.938, 0.871 and 0.188 for spur leaf area, crop load and their interaction, and from 0.864, 0.763 and 0.252 to 0.862, 0.755 and 0.245 for cane. Thus unequal cell counts did not change the ranking.

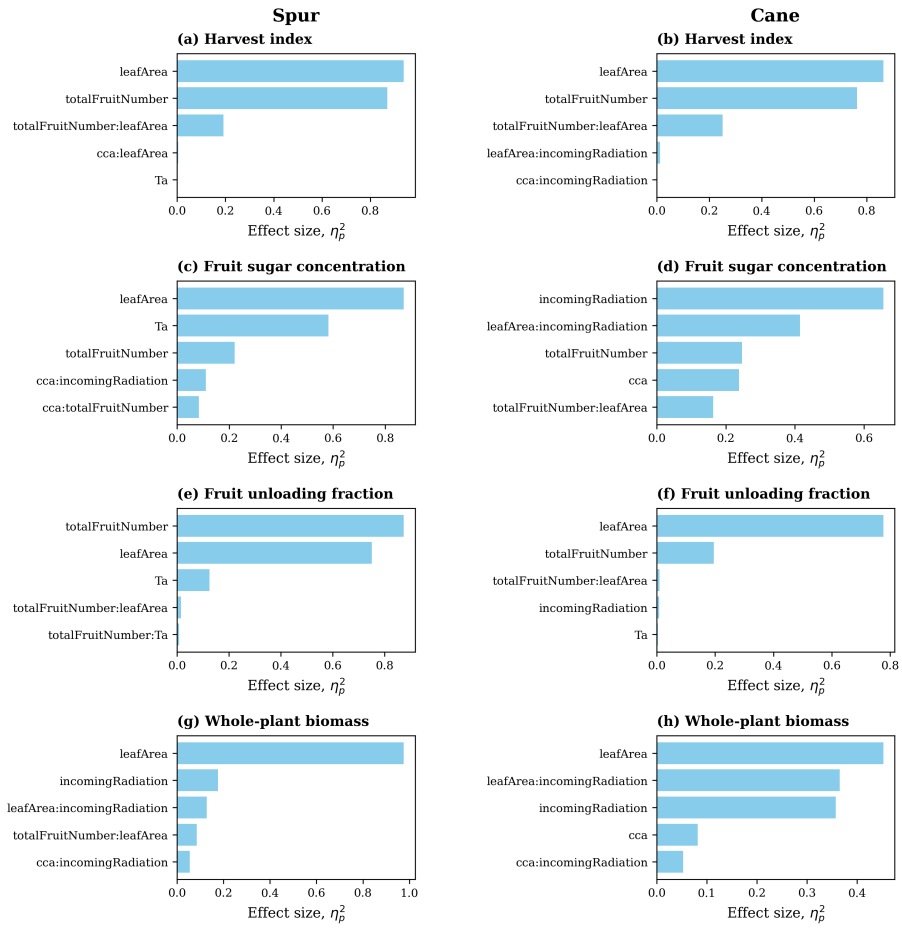


Figure S2: Leading partial η_p^2 terms for harvest index (A, B), fruit sugar concentration (C, D), fruit unloading fraction (E, F) and whole-plant biomass response (G, H) in retained spur (left) and cane (right) simulations. Axis labels retain the internal analysis-field names. Effect-size rankings are conditional on the fitted models and sampled factor ranges, and values are non-additive.

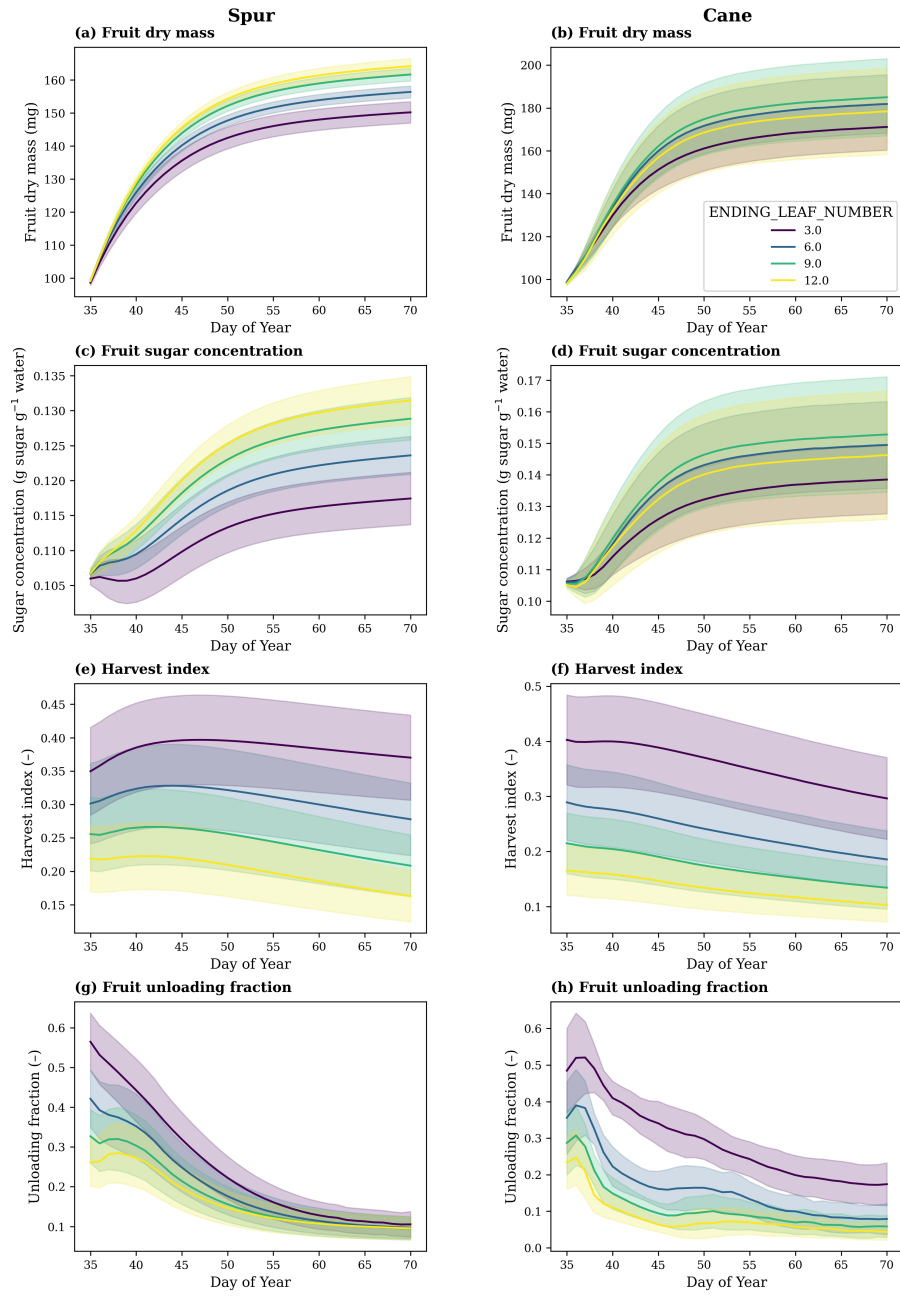


Figure S3: Daily trajectories of mean fruit dry mass (A, B), fruit sugar concentration (C, D), harvest index (E, F) and fruit unloading fraction (G, H) for spur (left) and cane (right) grapevines grouped by final leaf number. Lines show means; bands show mean ± 1 SD across retained crop-load, CO₂, radiation and temperature combinations at each day and leaf-number level. The bands represent dispersion among deterministic scenarios, not biological replication or uncertainty in the mean.

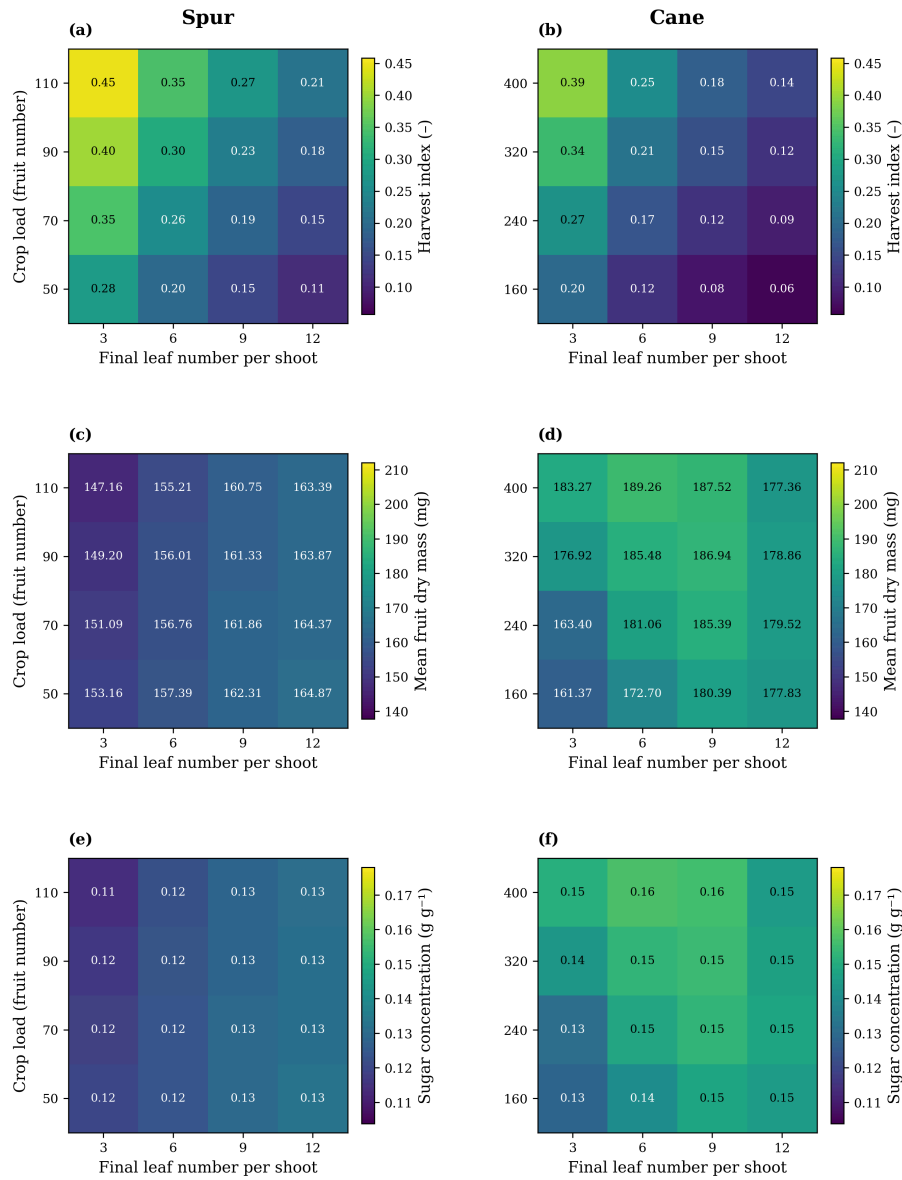


Figure S4: Leaf-number × crop-load response surfaces for harvest index (A, B), mean fruit dry mass (C, D; mg) and fruit sugar concentration (E, F) in spur (left) and cane (right) systems. Cells show means across retained CO₂, radiation and temperature combinations. Crop-load ranges were 50–110 fruits plant⁻¹ for spur and 160–400 fruits plant⁻¹ for cane.

5 Organ development framework

5.1 Development workflow architecture

FruitCropXL separates graph initiation, state updating and topology development. At simulation initiation, the configured initiation implementation constructs a plant graph or imports an existing architecture. During simulation, the update workflow advances environmental, physiological and organ states on existing objects, whereas the development implementation applies graph-rewrite rules when dynamic topology is enabled. Static configurations retain the initiated topology and update only geometry and state. Species- or scenario-specific

implementations can replace these workflows while retaining the same graph objects and process interfaces.

Development therefore exchanges state with, rather than operating independently of, the physiological calculations. Organ age and phenological state determine developmental eligibility; existing geometry supplies the basis for potential growth; accepted water potential and active phloem sugar concentration limit expansion; and carbon allocation or transport receives the resulting organ demands. Once allocation is accepted, organ geometry, structural carbon and reserve state are committed before later graph and aggregate updates.

5.2 Phenological state representation

Phenology can be disabled, prescribed through an initiated state, or advanced by the configured crop-specific implementation. Calendar and accumulated temperature variables provide developmental timing, while phenological and organ-age states gate events such as organ activation, growth windows, fruit development and management timing. These states are stored on the persistent graph or associated runtime components so that downstream physiological modules operate on the same accepted developmental context. The exact stage system, thresholds and transition rules belong to the selected crop implementation rather than to a universal FruitCropXL phenology algorithm.

5.3 Organ initiation and topology updates

Starting architecture can be generated by coded initiation rules, reconstructed from architecture inputs or created through structurally variable implementations. When dynamic development is active, bud and apex states are evaluated by the selected development implementation and eligible graph-rewrite rules add, replace or terminate biological objects. New organs are attached to explicit parent objects and inherit the plant identity and local topological context needed by radiation, hydraulic, carbon and output calculations. With static architecture, these topology-producing rules are suppressed while existing organs continue to update.

5.4 Organ growth demand and realised expansion

Growth-capable organs calculate potential geometric increments from developmental age or state, convert the corresponding area or volume increment into structural-carbon demand, and expose that demand to the active carbon mode. Accepted water and carbon responses determine realised primary expansion; eligible woody axes and structural roots can subsequently undergo secondary radial growth. Ageing, turnover, senescence and removal remain separate from this growth-demand calculation and from scheduled management removal.

5.4.1 Primary expansion

For leaves and young shoot axes, potential elongation is the derivative of the configured beta trajectory multiplied by the increment in developmental age:

$$\Delta L_o^{\text{pot}} = g_{\beta,o}(u_o) \Delta u_o, \quad (54)$$

$$\begin{aligned}
g_{\beta,o}(u) &= L_o^{\max} \frac{2t_{e,o} - t_{m,o}}{t_{e,o}(t_{e,o} - t_{m,o})} \\
&\times \left(\frac{t_{m,o}}{t_{e,o}} \right)^{t_{m,o}/(t_{e,o} - t_{m,o})} \frac{t_{e,o} - u}{t_{e,o} - t_{m,o}} \\
&\times \left(\frac{u}{t_{m,o}} \right)^{t_{m,o}/(t_{e,o} - t_{m,o})}, \quad 0 < u < t_{e,o}.
\end{aligned} \tag{55}$$

Here u_o and Δu_o are organ age and its hourly increment in the configured age domain, $L_o^{\max}$ is potential final length (m), and $t_{m,o}$ and $t_{e,o}$ are ages in that same domain at maximum growth rate and completion. Ratios and exponents of age are dimensionless; $g_{\beta,o}$ is length per unit developmental age and ΔL_o^{pot} is the potential length increment (m per hourly step). The rate is zero outside the active interval. The symbol u_o avoids confusion with the graph-resolved carbon-transport intercept a_o .

The hydraulic response of primary expansion is

$$f_{w,o} = \text{clip} \left(\frac{1}{1 + \exp[a_{w,o}(\psi_o - b_{w,o})]}, 0, 1 \right), \tag{56}$$

where ψ_o is accepted organ water potential (MPa), $b_{w,o}$ is its configured half-response potential (MPa), and $a_{w,o}$ has units MPa^{-1} , making the exponential argument dimensionless. The clipping operation and response $f_{w,o}$ are dimensionless. Carbon response $f_{c,o}$ is also dimensionless and uses Section 2.4 with $x_o = C_{\text{pool}}$ in common-pool carbon allocation or $x_o = c_{\text{sug},o}$ in graph-resolved carbon transport. Leaves and young annual internodes use the dominant limitation rather than multiplying the two stresses:

$$\Delta L_l = f_{Mb,\text{shoot}} \Delta L_l^{\text{pot}} \min(f_{w,l}, f_{c,l}), \quad \Delta L_i = \Delta L_i^{\text{pot}} \min(f_{w,i}, f_{c,i}). \tag{57}$$

Here $f_{Mb,\text{shoot}}$ is the dimensionless plant shoot-root balance multiplier; ΔL_l and ΔL_i are accepted realised length increments (m per hourly step), whereas superscript **pot** denotes the corresponding potential increments. Young-internode early radial expansion uses the same dimensionless $\min(f_w, f_c)$ response. In temperature-only growth configurations, $f_w = 1$.

Structural-carbon demand is calculated from realised geometry. Representative leaf-area and cylindrical-axis expressions are

$$D_l^{\text{pri}} = f_C q_g \rho_A \Delta A_l, \tag{58}$$

$$D_i^{\text{pri}} = f_C q_g \rho_i \pi (r_i^2 \Delta L_i + 2r_i L_i \Delta r_i^{\text{early}}), \tag{59}$$

where D_l^{pri} and D_i^{pri} are realised primary structural demands (mg C per hourly step), and f_C and q_g are the dimensionless dry-matter carbon fraction and growth-cost multiplier. ρ_A is leaf dry mass per area (mg dry matter m^{-2}), and ΔA_l is realised polygon-area increment (m^2 per step). For the cylindrical axis, r_i , L_i , ΔL_i and $\Delta r_i^{\text{early}}$ are in m, while ρ_i is dry-matter density (mg dry matter m^{-3} after the implementation's geometric conversion); the bracketed term is a volume increment (m^3 per step). Potential demands use the same expressions with potential geometric increments, and realised demands use accepted increments after the dimensionless

water and carbon responses. Other organ implementations use the corresponding area or volume increment.

Fine-root demand uses a relative-growth term bounded by capacity:

$$D_r^{\text{pri}} = \min[C_r^{\text{str}} q_g k_{\text{RGR}} f_T \min(f_{w,r}, f_{c,r}) f_{Mb,r}, (M_r^{\text{max}} - M_r) f_C], \quad (60)$$

where D_r^{pri} is realised fine-root structural demand (mg C per hourly step), C_r^{str} is fine-root structural carbon (mg C), f_T , $f_{w,r}$, $f_{c,r}$ and $f_{Mb,r}$ are dimensionless temperature, water, carbon and shoot-root-balance responses, and $(M_r^{\text{max}} - M_r) f_C$ is the remaining carbon capacity (mg C). The implementation multiplies the configured relative-growth parameter by the hourly timestep; its resulting k_{RGR} is therefore a dimensionless increment per step in this displayed equation. The potential demand uses the water- and temperature-limited expression before the carbon response, whereas the displayed realised demand additionally applies $\min(f_{w,r}, f_{c,r})$ and the capacity bound. Root exudation and turnover are evaluated separately.

5.4.2 Secondary growth

Eligible annual internodes, perennial axes and structural roots calculate a potential radial increment from the configured base rate, temperature, seasonal position and organ age:

$$\Delta r_o^{\text{base}} = k_{\text{sec},o} f_T(T_a) f_{\text{season},o}(d) f_{\text{age},o}(a_y), \quad (61)$$

Here $k_{\text{sec},o}$ is potential radial growth (m h^{-1}); T_a is air temperature ($^{\circ}\text{C}$), d is seasonal time (days), and a_y is organ age (years). The response functions f_T , $f_{\text{season},o}$ and $f_{\text{age},o}$ are dimensionless, so Δr_o^{base} is a radial increment in m over the hourly step. Their exponential arguments, where used, are ratios to configured scales and are dimensionless. Annual internodes instead use their post-elongation cambial window. Demand-forming and realised increments are

$$\Delta r_o^{\text{dem}} = \Delta r_o^{\text{base}} f_o^{\text{dayC}}, \quad \Delta r_o^{\text{real}} = \Delta r_o^{\text{dem}} f_{\text{sink},o,\text{sec}}, \quad (62)$$

where f_o^{dayC} is the dimensionless previous-day carbon-status scale where active and $f_{\text{sink},o,\text{sec}}$ is the dimensionless secondary-growth sugar response. Δr_o^{dem} and Δr_o^{real} are potential demand-forming and accepted realised radial increments (m per hourly step). Structural roots omit the daily-carbon multiplier. Secondary growth has no direct water-potential multiplier; only young-internode early radial expansion uses the primary water response. Potential and realised carbon demand use the first-order cylindrical-shell increment:

$$D_o^{\text{sec,pot}} = f_C q_g 2\pi r_o \ell_o \rho_o \Delta r_o^{\text{dem}}, \quad D_o^{\text{sec,real}} = f_C q_g 2\pi r_o \ell_o \rho_o \Delta r_o^{\text{real}}, \quad (63)$$

Here $D_o^{\text{sec,pot}}$ and $D_o^{\text{sec,real}}$ are potential and realised secondary structural-carbon demands (mg C per hourly step); r_o and ℓ_o are current radius and length (m), ρ_o is dry-matter density (mg dry matter m^{-3}), and f_C and q_g are dimensionless. Thus the shell-volume increment multiplied by density, carbon fraction and growth cost gives mg C per step; ℓ_o is used for geometric length to distinguish it from source loading L_o .

Secondary growth competes with primary growth, maintenance and reserve storage in the same carbon solve. Committed diameter increases by $2\Delta r_o^{\text{real}}$; geometry-specific post-commit

updates preserve the fixed length of above-ground perennial axes and reconcile structural-root geometry with accepted biomass.

5.4.3 State commitment and growth costs

Primary and secondary growth use the common growth-cost coefficient q_g when converting new tissue production into structural-carbon demand. The resulting structural demand therefore includes the carbon cost associated with growth before it enters the carbon allocation or transport calculation. Accepted structural unloading is counted once as an organ sink, and growth respiration is not introduced as an additional carbon demand during allocation.

At state commitment, accepted structural unloading is partitioned between growth respiration and carbon retained in new structural biomass:

$$R_{g,o} = D_o^{\text{str}}(q_g - 1), \quad \Delta C_o^{\text{str}} = D_o^{\text{str}} - R_{g,o} = D_o^{\text{str}}(2 - q_g), \quad (64)$$

where D_o^{str} is accepted structural unloading (mg C per hourly step), q_g is the dimensionless growth-cost coefficient, $R_{g,o}$ is carbon consumed by growth respiration, and ΔC_o^{str} is the carbon retained as new structural biomass (mg C per hourly step). The same growth-cost formulation is applied across growth-capable plant organs, while organ-specific growth equations determine the underlying potential structural demand. Reported states include potential and realised geometric increments, water and carbon response factors, structural demand, growth respiration, structural carbon and biomass.

5.5 Management operations

Management operations are implemented as scheduled planners that query and modify the existing graph. Pruning, leaf removal and fruit thinning remove or deactivate selected organs, after which the active update and development workflows continue from the modified topology and state. Management rules can use calendar, phenological, spatial or organ-state criteria, but their crop- and experiment-specific selection logic is configured separately from the shared graph-editing workflow. Removed biomass is recorded only by operations implemented as management events.

5.6 State continuation and planner modules

Configured plant graphs and their retained organ states can be saved, reloaded and continued without replaying the complete developmental history. A restored state can therefore provide a common initial graph for alternative weather, soil-water, crop-load or management scenarios. Planner modules can also align bud, leaf or fruit populations with configured targets while retaining compatible existing objects where possible.

In the apple crop-load demonstration, fruit populations are aligned using grid-level fruit twinning. The canopy is divided into a two-dimensional management grid defined by local along-row position and height, and each flower or fruit is assigned to a grid cell. At the configured T1 or T2 stage, current counts are reconciled with cell-specific targets. Depending on configuration, deficits can be filled by adding fruits at eligible fruiting sites, subject to whole-tree and site-level limits, whereas excess fruits are removed within each cell, with smaller organs removed first. Twinning is treated as state alignment rather than explicit pruning; separate pruning events account for removed biomass in management outputs.

The initiation, phenology, bud, apex and organ-creation rules are supplied by the configured development implementation. Crop-specific stage definitions, thresholds and architectural algorithms are outside the scope of this framework description and can be documented in dedicated crop-model descriptions where required. Static and reduced configurations use the same process interfaces but may omit dynamic topology development.

6 Consolidated notation for the process equations

The notation below consolidates the principal symbols used in the main and supplementary radiation, gas-exchange, hydraulic, carbon, growth and fruit equations. Indices i and j denote connected transport states or segment endpoints, l a leaf, o an organ, p a process and r a root compartment; PB denotes PlantBase. Superscripts `pot`, `real` or `act`, `trial`, `raw`, `str`, `res` and $t - 1$ identify potential, accepted realised, solver-candidate, pre-correction, structural-carbon, reserve-carbon and preceding accepted hourly states, respectively; superscript 0 is the current linearisation point and a prime denotes a coefficient transformed to the upstream segment endpoint. Carbon concentrations and fluxes retain the different units used by common-pool carbon allocation and graph-resolved carbon transport.

Table S7: Consolidated notation for the principal equations in the main and Supplementary Methods.

Symbol	Definition	Unit
α_s, θ_s, h_s	Solar azimuth, zenith and elevation angles; $h_s = 90^\circ - \theta_s$	degree
E_b	Beam-normal clear-sky radiation	W m^{-2}
I_g, I_b, I_d	Horizontal global, direct and diffuse radiation	W m^{-2}
k_t, f_d	Clearness index and diffuse fraction	dimensionless
r_l	Relative incident PAR at leaf l	dimensionless
$C_{\text{top}}, k, L_{\text{above},l}$	Canopy-top relative PAR, extinction coefficient and leaf-area index above leaf l	dimensionless
$I_{\text{inc},l}$	Incident PAR at leaf l	$\mu\text{mol photons m}^{-2} \text{s}^{-1}$
$A_{n,c,l}, A_{n,j,l}$	Rubisco- and electron-transport-limited net assimilation	$\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$
$A_{n,l}^{\text{pot}}, A_{n,l}^{\text{lim}}, A_{n,l}^{\text{act}}$	Potential, temperature-updated limiting and hydraulically accepted net assimilation	$\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$
$C_{c,l}, C_{i,l}, \Gamma_l^*$	Chloroplast and intercellular CO_2 partial pressure and photorespiratory compensation point	μbar
$V_{\text{cmax},l}, R_{d,l}$	Maximum Rubisco carboxylation capacity and day respiration	$\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$
$K_{c,l}, K_{o,l}, O$	Michaelis constants for CO_2 and O_2 , and oxygen partial pressure	μbar for K_c ; mbar for K_o and O
$J_l, J_{\text{max},l}$	Realised and maximum electron-transport rates	$\mu\text{mol electrons m}^{-2} \text{s}^{-1}$

Table S7 (continued): gas exchange and hydraulics.

Symbol	Definition	Unit
$\kappa_{2,l}, \theta$	Low-light conversion efficiency and non-rectangular-hyperbola curvature	mol electrons mol ⁻¹ photons; dimensionless
$g_{s,c,l}, g_{s,w,l}, g_0$	Stomatal conductance to CO ₂ , stomatal conductance to water vapour and residual conductance	mol m ⁻² s ⁻¹
$g_{s,w,l}^{(p)}$	Physical stomatal conductance to water vapour	m s ⁻¹
D_a, e_s, e_a	Atmospheric vapour-pressure deficit, saturation vapour pressure and ambient vapour pressure	kPa
E_l	Leaf transpiration per unit leaf area	mm s ⁻¹ , numerically kg water m ⁻² s ⁻¹
$R_{n,l}, Q_{\text{abs},l}, R_{\text{lw},l}$	Net, absorbed short-wave and signed net long-wave radiation	J m ⁻² s ⁻¹
T_l, T_a	Leaf and air temperature	°C
$S(T_l, T_a), \Delta(T_a)$	Secant and tangent slopes of saturation vapour pressure	kPa K ⁻¹
$\lambda, \rho c_p, \gamma$	Latent heat of vaporisation, volumetric heat capacity of air and psychrometric constant	J kg ⁻¹ ; J m ⁻³ K ⁻¹ ; kPa K ⁻¹
$r_{a,l}, r_{s,l}, r_{bw,l}, r_{t,l}$	Aerodynamic, stomatal, boundary-layer and turbulence resistances	s m ⁻¹

Table S7 (continued): hydraulic transport.

Symbol	Definition	Unit
PLC_l	Percentage loss of leaf hydraulic conductivity expressed as a fraction	dimensionless
$\psi_l, \psi_i, \psi_j, \psi_{rx,r}$	Leaf, above-ground segment-end and fine-root-compartment xylem water potentials	MPa
$\psi_{PB} = \sum_r f_r \psi_{rx,r}$	Uptake-fraction-weighted PlantBase xylem water potential and root-shoot hydraulic boundary	MPa
$[ABA]_x$	Xylem ABA concentration	$\mu\text{mol m}^{-3}$
F_{ij}, F_l, F_w	Segment, leaf and whole-plant water fluxes	mg water s^{-1}
K_{ij}, K_i, K_l	Segment, internode and leaf hydraulic conductance	$\text{mg water s}^{-1} \text{MPa}^{-1}$
$R_w(F_w)$	Difference between trial whole-plant water supply and summed realised leaf flux	mg water s^{-1}
d_i, L_i, ℓ_{ij}	Segment diameter and stored transport lengths	m
$c_{x,1}, c_{x,2}, c_{x,3}, \phi_{\text{stem}}$	Vulnerability-curve exponent, empirical maximum-conductance coefficient, diameter exponent and vulnerability-potential scale	dimensionless; dimensionless; dimensionless; MPa
$G_{xl,0}, G_{xl,\min}, G_{xl,\max}, K_{l,\min}$	Baseline, raw lower/upper and effective lower leaf-conductance bounds	$\text{mg water s}^{-1} \text{MPa}^{-1}$
$\tau_{tr,l}, \tau_{ABA,l}$	Transpiration and ABA leaf-conductance response coefficients	MPa^{-1} ; dimensionless
D_w, g	Implemented water-density constant and gravitational acceleration	g cm^{-3} ; m s^{-2}

Table S7 (continued): carbon allocation, transport and growth.

Symbol	Definition	Unit
$C_o^{\text{str}}, C_o^{\text{res}}$	Organ structural and reserve carbon	mg C
$S_{A,l}, D_{o,p}^{\text{pot}}$	Assimilation supply and potential process demand accumulated over the step	mg C per hourly step
$L_o^{\text{pot}}, L_o, U_{o,p}$	Potential loading, realised loading and realised sink unloading in common-pool bookkeeping	mg C per hourly step
Q_o^{leak}	Passive carbon leakage into organ reserve	mg C per hourly step
C_{pool}	Spatially uniform common-pool sugar concentration, represented as a regulatory sugar mass fraction and stored on PlantBase	g sugar g ⁻¹ phloem sap
$c_{C,o}, c_{\text{sug},o}$	Graph-local phloem carbon and converted sugar concentrations	g C cm ⁻³ ; g sugar cm ⁻³ phloem sap
$\Phi_o = c_{C,o}^2/2$	Graph-local carbon potential	(g C cm ⁻³) ²
ρ_{sap}	Phloem-sap density relating sugar mass fraction to volumetric sugar concentration	g phloem sap cm ⁻³
j_o	Signed graph-resolved carbon-transport organ exchange; positive to sinks and negative from sources	g C h ⁻¹
a_o, b_o	Intercept and slope of the local $j_o - \Phi_o$ linearisation	g C h ⁻¹ ; g C h ⁻¹ per unit Φ
$\beta_{\text{res}}(d); d, t_{m,r}, t_{e,r}$	Reserve-season interpolation state; elapsed time from the seasonal origin, maximum-rate time and transition-completion time	dimensionless; days
$K_{m,o,\text{res}}(d); K_{m,o}^{\text{early}}, K_{m,o}^{\text{late}}$	Active reserve-synthesis half-response concentration and its configured early- and late-season endpoints	active concentration unit: g sugar g ⁻¹ phloem sap or g sugar cm ⁻³ phloem sap
$k_{\text{hyd},o}(d); k_{\text{hyd},o}^{\text{early}}, k_{\text{hyd},o}^{\text{late}}$	Effective reserve-hydrolysis coefficient and its configured early- and late-season endpoints	h ⁻¹
$k_{\text{syn},o}$	Reserve-synthesis coefficient, independent of the reserve-season interpolation	h ⁻¹

Table S7 (continued): folded carbon transport and growth.

Symbol	Definition	Unit
A, B, A', B'	Node-aggregated and upstream-folded transport coefficients	same respective units as a_o and b_o
r, x_p	Segment phloem resistance and non-negative xylem-coupling decrement	Φ per (g C h ⁻¹); Φ
$\Phi_{PB}^{raw}, \Phi_{PB}, A_{PB}, B_{PB}$	Raw zero-net-flow candidate, accepted PlantBase carbon potential, intercept and slope	carbon potential; carbon potential; g C h ⁻¹ ; g C h ⁻¹ per unit carbon potential
$c_{sug,PB} = \kappa_s \sqrt{2\Phi_{PB}}$	PlantBase phloem sugar concentration converted from the graph boundary	g sugar cm ⁻³ phloem sap
ε_C	Sum of absolute changes in organ exchange between graph-resolved carbon-transport iterations	g C h ⁻¹
$K_{m,o,p}$	Process half-response concentration	active concentration unit: g sugar g ⁻¹ phloem sap or g sugar cm ⁻³ phloem sap
$\beta_o, C_{0,o}$	Source-loading slope and half-response concentration	inverse active concentration unit; active concentration unit
R_C	Common-pool loading-minus-leakage-minus-unloading residual	mg C per hourly step
$R_{g,o}, \Delta C_o^{str}$	Growth cost and committed structural-carbon increment	mg C per hourly step

Table S7 (continued): fruit water and sugar relations.

Symbol	Definition	Unit
F_x, F_p	Fruit xylem and phloem water influx	g water h ⁻¹ after FruitCropXL synchronisation; backend-native internally
A_i, L_i, σ_i	Fruit vascular exchange area, area-specific hydraulic conductivity and reflection coefficient	BerryComplex : cm ² ; g water cm ⁻² h ⁻¹ bar ⁻¹ ; dimensionless. VirtualFruit : backend-native.
$P_i, P_f, \Pi_i, \Pi_f, \psi_f$	Vascular and fruit hydrostatic, osmotic and water potentials	bar in BerryComplex ; backend-specific in VirtualFruit
U_f, U_a, U_m, U_d	Total, active, mass-flow and optional diffusive fruit sugar import	g sugar h ⁻¹ after synchronisation; backend-native internally
$C_p, C_f, K_{m,f}$	Phloem sugar concentration, fruit sugar concentration and uptake half-response concentration	C_p : active phloem basis (g sugar g ⁻¹ phloem sap or g sugar cm ⁻³ phloem sap); C_f : g sugar g ⁻¹ fruit water; $K_{m,f}$ follows the phloem basis
$M_{d,f}, M_{w,f}$	Fruit dry and water mass	mg after synchronisation; backend-native internally
W_{resp}, T_f	Respiration-produced water and fruit transpiration	g water h ⁻¹ after synchronisation; backend-native internally

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